

Final Exercise

Submission deadline: July 10, 2026 (August 13 for special circumstances acknowledged by BGU)

Critical! AI Use Policy

Artificial intelligence tools (e.g. ChatGPT, Claude, Copilot, Gemini, etc.) may be used **only in limited ways** that support learning and technical clarification. They may not replace the intellectual work being evaluated in this project.

You are responsible for every part of your submission. You must be able to independently explain, justify, and reproduce every aspect of the submitted work: algorithms, implementation details, parameter choices, numerical behavior, and interpretation of results. You will be personally interviewed on these aspects.

Permitted uses of AI tools: Conceptual clarification of terminology, explanation of background theory, limited debugging assistance, and minor syntax or documentation help. You may also use AI tools for grammar and writing clarity. Examples of permitted use:

“Explain the meaning of the factors in the SVD.”

“Why does this NumPy broadcasting error occur?”

“How is the asymptotic complexity of an algorithm defined?”

Prohibited uses of AI tools: generating substantial portions of code unless instructed otherwise, design of algorithms or software architecture, interpretation or analysis of any results, generating conclusions or discussion sections, writing of substantial technical discussion, completion of intellectual work of the assignment. Examples of prohibited use:

“Compute the trends in the plot for me.”

“Interpret these convergence plots.”

“Analyze the behavior in these graphs and summarize the conclusions for me.”

You may not submit code, derivations, explanations, or analysis that you do not personally understand.

Required AI Disclosure:

If you use AI tools in any capacity, add an appendix to your report stating:

- AI tool(s) used
- Purpose of use
- Example prompts and links to documented conversations with AI
- Brief description of how AI outputs were independently verified

Failure to disclose significant AI use may be treated as academic misconduct.

Guiding Principle: AI may support learning, but it may not perform the engineering, computational reasoning, analysis, or scientific interpretation being evaluated in this project.

Important! How to Not Lose Points on Style and Presentation

Follow these guidelines to make it easier to grade your work favorably: (1) Use clear idiomatic English. (2) Use consistent fonts for all notations. Use distinct fonts for vectors, matrices, scalars, and non-equation text. (3) Make your plots clear and informative. Use legends, axis titles, and units. (4) Make sure the data in the figure do not cluster or lose meaning – distribute data evenly and informatively within the figures: use *log-log* or *semi-log* scales wherever wide dynamic ranges of values are of interest, especially when values approaching zero are presented. (5) Use “logspace”, rather than “linspace”, when looping over a broad dynamic ranges of values. (6) *Do not* connect discrete data points with lines – this introduces misleading visual information. (7) Instead, add trendlines and asymptotes for showing scaling. (8) When comparing plots, put them *on the same figure*. Do not leave it to the reader to visually compare figures placed side-by-side or, worse, in separate pages or with different value ranges. (9) Do not just describe your results! Explain and interpret them. (10) Lastly, do not include any code in your report!

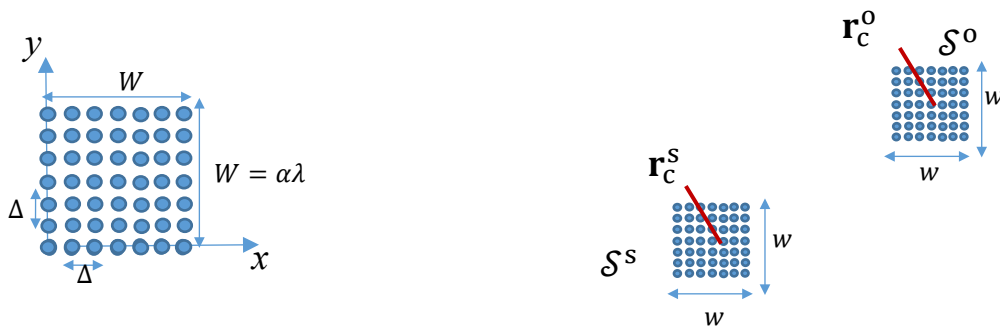
Use your personal ID number to determine your tasks:

- Rightmost digit (for singles): odd (£), even (££). Couples do both.
- Second from the right digit (for singles): odd (¥), even (¥¥). Couples do both.
- Third from the right digit (for singles): odd (\$), even (\$\$). Couples do both.
- Fourth from the right digit (for singles): odd (€), even (€€). Couples do both.

Background

This exercise studies the ability of various algebraic structures to represent efficiently a many-body wave problem of interactions between wave scattering particles

Consider a squared region $S \in \mathbb{R}^2$ of dimensions $W \times W$, where $W = \alpha\lambda$, centered at the origin. In the square, particles defining a set $\mathcal{S}$ are placed on a uniform cartesian grid, with a spacing of $\Delta = \lambda/10$ in each dimension between them. We denote $\mathbf{r}_m = (x_m, y_m)$ the location of the m th particle.



Each particle is influenced by itself and all other particles. The matrix describing the interactions between the particles is denoted A and its entries are given by

$$A_{mn} = K(\mathbf{r}_m - \mathbf{r}_n)$$

In this exercise, we will examine the ability to compress A using various representations and its dependence on the interaction as represented by the function K .

Part I – Preliminaries

Let $S^s \in S$ and $S^o \in S$ be two subsets of particles. The subsets are non-overlapping and are confined to squared subdomains $S^s \in S$ and $S^o \in S$ that are of dimensions $w \times w$ and are centered at $\mathbf{r}_c^s$ and $\mathbf{r}_c^o$. The block A^{os} of A is composed of the interactions between the two sets.

- (a) Implement a routine that, provided w , $\mathbf{r}_c^s$, and $\mathbf{r}_c^o$ (and the other problem parameters) constructs the blocks A^{os} . Use the kernel function

$$K(\mathbf{r}_m - \mathbf{r}_n) = \begin{cases} \frac{1}{|\mathbf{r}_m - \mathbf{r}_n|^{1/2}} & m \neq n \\ \frac{1}{2} & m = n \end{cases}$$

For the demonstrations, pick $\mathbf{r}_c^s$ and $\mathbf{r}_c^o$ such that $S^s \in S$ and $S^o \in S$ are two remote corners of A^{os} , for the parameters $\lambda = 1$, $\alpha = 4$, $w = W/8$. Compute also the singular value decomposition (SVD) of A^{os} . Plot A^{os} and its singular values curve. Discuss the structure of the matrix and singular value curve.

- (b) Scale the problem in (a) by repeatedly doubling W (continue until A^{os} is too large to handle). Show, in a single figure, a sequence of singular value curves. Plot the rank $\mathcal{R}(\tau)$ as a function of W for the following relative error thresholds: $\tau = 10^{-2}, 10^{-5}, 10^{-8}$. Also plot the computation time as a function of W . Explain the results.
- (c) Repeat (b) for the kernel

$$K(\mathbf{r}_m - \mathbf{r}_n) = \begin{cases} \frac{e^{-jk|\mathbf{r}_m - \mathbf{r}_n|}}{|\mathbf{r}_m - \mathbf{r}_n|^{1/2}} & m \neq n \\ \frac{1}{2} & m = n \end{cases}$$

where $k = 2\pi/\lambda$. Describe the results and the difference in the behavior of the ranks from that in (b).

- (d) **Use an AI chat platform for this question:** (£) use Gemini. (££) use Copilot. *Couples do both* and discuss differences in the outputs.

In your own words and without graphics or mathematical formulas whatsoever, describe to the machine the nature of the experiments conducted in (b) and (c) and the difference you observed between their results. Ask the chat for possible explanations for what you observed. In your report, include the prompt you used, a link to the chat, a concise summary in your own words of the AI's response, your own critical evaluation of the machine's answer (and, for couples, a critical discussion of differences between machines).

Part II – Fast Rank Estimation

The following algorithm is proposed for fast estimation of the rank $\mathcal{R}(\tau)$ of A^{os} .

1. Set $l := 1$, $n := 10$.
2. Randomly draw two sets of indices $\mathbf{i}_{\text{row}} \in \mathbb{Z}^n$ and $\mathbf{i}_{\text{col}} \in \mathbb{Z}^n$ such that each vector contains a set of non-repeating indices in the range $[1, N]$.
3. Extract the submatrix $A_l^{os} := A^{os}(\mathbf{i}_{\text{row}}, \mathbf{i}_{\text{col}})$.
4. Compute the SVD of A_l^{os} and its rank $\mathcal{R}_l(\tau)$.
5. If $l \neq 1$, compute $\varepsilon_{\mathcal{R}} = (\mathcal{R}_l - \mathcal{R}_{l-1})/\mathcal{R}_l$. If $\varepsilon_{\mathcal{R}} < \tau_{\mathcal{R}}$, terminate and return $\mathcal{R}_l$.
6. Update $l := l + 1$, $n := 2n$, and go to Step 2.

- (e) Explain the operation of the algorithm. Specifically, discuss the roles of $\varepsilon_{\mathcal{R}}$ and $\tau_{\mathcal{R}}$. What should be required of A^{os} for the algorithm to work?
- (f) Repeat the rank study in (c) with the proposed algorithm. Compare its rank predictions to the SVD ranks. Push for larger W values. How far can you go on the same machine used in (c)? Find the asymptotic scaling of the rank with W .
- (g) **Use an AI chat platform for this question:** (¥) use Claude. (¥¥) use ChatGPT. *Couples do both* and discuss differences in the outputs.
In your own words and without graphics or mathematical formulas, describe to the chat the nature of the plot produced in (f) for the oscillatory kernel in (c). Ask the chat to explain the reason for the rank scaling the way it does and summarize the answer in your own words. In your report, include the prompt you used, a link to the chat, a concise summary in your own words of the AI's response, your own critical evaluation of the machine's answer (and, for couples, a critical discussion of differences between machines).
- (h) Plot the computation time as a function of W and find the asymptotic scaling of the computation time with W . Explain the results and whether they match the theory. How do you expect the complexity to change for the kernel in (a)?

Part III – Fast LR Approximation

- (i) Review and explain the operation of one of the algorithms below (*couples do both*), for fast LR approximation of A^{os} for a given τ . In particular, discuss the algorithm parameters (e.g., γ, N_b) and their potential influence on its accuracy/convergence. Analyze the computational complexity.

(¥) Alg. Informed Randomized LR approximation

1. Compute an approximation $\mathcal{R}_l$ of the rank using the algorithm in Part II.
2. Construct a matrix $G \in \mathbb{C}^{N \times (\mathcal{R}_l + \gamma)}$ of complex Gaussian i.i.d. random values.
3. Compute $M := AG$.
4. Compute $U\Sigma V^* = M$ using the SVD.
5. Use $\hat{U} := U(:, 1: \mathcal{R}_l)$ and compute $\hat{B} = \hat{U}^* A$.
6. Return the factors $\hat{U}$ and $\hat{B}$ (that form $\hat{U}\hat{B} \approx A$).

(¥¥) Alg. Informed Blocked Randomized LR approximation

1. Compute an approximation $\mathcal{R}_l$ of the rank using the algorithm in Part II.
2. Set $\mathcal{B}_0 = \mathcal{R}_l / N_b$ ($N_b \in \mathbb{Z}$). Set $\hat{U} = [], \hat{B} = []$. Compute $\|A\|_F$. Set $l = 1$
3. Construct a matrix $G \in \mathbb{C}^{N \times (\mathcal{B}_0 + \gamma)}$ of complex Gaussian i.i.d. random values.
4. Compute $M := AG$.
5. Compute $U\Sigma V^* = M$ using the SVD.
6. $B := U^*(:, 1: \mathcal{B}_0)A$
7. Update $\hat{U}: [\hat{U} \ U(:, 1: \mathcal{B}_0)], \hat{B}: [\hat{B} \ B]$;
8. If $\|B\|_F \leq \tau \|A\|_F$, terminate and return the factors $\hat{U}$ and $\hat{B}$ (that form $\hat{U}\hat{B} \approx A$) and $\mathcal{R}_l = l\mathcal{B}_0$.
9. Update $A := A - U(:, 1: \mathcal{B}_0)B$, $l := l + 1$
10. Go to step 2.

- (j) Implement the algorithm(s). Apply it for the geometric configuration, kernel, and thresholds in (c). Specifically, examine the computation time scaling compared to that of a straightforward SVD, and the relative difference between the fast LR approximation and the best one achieved using the SVD. Use this test case to suggest values for the algorithm parameters that are sufficient for its operation.

- (k) Use the algorithm with the parameters you found and study the scaling of its computation time with N in the same manner as in (b) and (c). Plot the computation times and ranks and compare these results to those of the SVD in (b) and (c). How does the complexity to change for the kernel in (b)?

Part IV – Hierarchical Matrix Compression

In this part, we wish to construct a hierarchical structure of blocks that will replace the entire A . In the structure, also termed an $\mathcal{H}$ -matrix, most of the off-diagonal blocks A^{os} are expressed in low-rank compressed manner. The blocks that are compressed are those identified as good candidates for compression by an “admissibility” recitation. For example, blocks that represent the interactions between sets of points that are sufficiently distant compared to their size are often compressed.

- (l) **Use your favorite AI chat tool** to learn about hierarchical matrices: Ask the chat to explain to you the principles of this compression approach. Guide the chat to discuss the reasoning behind this representation of matrices and the difference between strong and weak admissibility criteria. Summarize the conversation in your own words. Based on your study in (b) and (c), do you expect the method to be effective for either or both of these kernels?
- (m) **Use the AI chat** to build a data structure that represents a quadratic hierarchical tree of the points in $\mathcal{S}$, partitioned based on their geometric location within a hierarchy of squared domains. Based on the tree structure, ask the chat for a routine that builds a hierarchical matrix data structure in the form of a list of hierarchical matrix blocks. Ask the chat to use either (€) a “strong admissibility criterion” (€€) “weak admissibility criterion” (couple do both and compare) (ask the chat for an explanation of these if it did not come up in your conversation yet). For each block, a list entry will include: the participating row and column indices, their numbers, and an indicator of whether the block is admissible or not. Report the prompts used, the measures you took for verifying and correcting the results from the chat, and comment on the experience and quality of the responses you received.
- (n) Write (**not using machine anymore**) a code that goes over all the blocks in the data structure(s) from step (m) and computes the memory required for storing the entire $\mathcal{H}$ -matrix (including all admissible and inadmissible blocks), for given parameters of the $\mathcal{H}$ -matrix structure (e.g., depth of hierarchy, compression threshold τ , etc.). Use this code and show the scaling of the memory with W for various parameter choices, for the kernels in both (b) and (c). Discuss the savings compared to storage of the original A , the differences in the scaling of the memory between the two kernels, and the preferable choice of parameters for these two cases.

Good luck!